King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam II Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. A linear homogeneous constant-coefficient equation whose general solution is

$$y(x) = Ae^{2x} + B\cos(2x) + C\sin(2x)$$

is

(a) $y''' - 3y'' + 4y' - 5y = 0$

(b) $y''' - 4y'' + 5y' - 8y = 0$

(c) $y''' - 5y'' + 4y' + 8y = 0$

(d) $y''' - 2y'' + 4y' - 8y = 0$

(e) $y''' + 3y'' - 4y' + 5y = 0$

2. The values of k for which the three vectors

$$v_1 = (1, 0, 1), \quad v_2 = (2, -3, 4), \quad v_3 = (3, 5, k)$$

form a basis for $\mathbb{R}^3$ are

(a) $k \neq 1$

(b) $k \neq 0$

(c) $k \neq 5$

(d) $k \neq -\dfrac{1}{3}$

(e) $k \neq 2$

3. The Wronskian of the functions

$$f(x) = 1, \quad g(x) = x, \quad h(x) = x^2$$

is

(a) 2

(b) $2x$

(c) x^2

(d) 0

(e) 4

4. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$y''' - y'' - 12y' = x - 2xe^{-3x}$$

is given by

(a) $y_p(x) = Ax + Bx^2 + (Cx + Dx^2)e^{-3x}$

(b) $y_p(x) = A + Bx + (C + Dx^2)e^{-3x}$

(c) $y_p(x) = A + Bx^2 + (Cx + D)e^{-3x}$

(d) $y_p(x) = A + Bx + (Cx + Dx^2)e^{-3x}$

(e) $y_p(x) = Ax + Bx^2 + (C + Dx^2)e^{-3x}$

5. Let $y(x)$ be the solution of the initial value problem

$$y'' - 4y' + 3y = 0; \quad y(0) = 7,\ y'(0) = 11.$$

Then $y(\ln 2) =$

(a) 22

(b) 26

(c) 18

(d) 24

(e) 20

6. The rank of the matrix $A = \begin{bmatrix} 1 & 3 & 3 & 9 \\ 2 & 7 & 4 & 8 \\ 2 & 7 & 5 & 12 \\ 2 & 8 & 3 & 2 \end{bmatrix}$ is

(a) 0

(b) 3

(c) 1

(d) 2

(e) 4

7. Given that vectors $\vec{t} = (2, -7, 9)$, $\vec{u} = (1, -2, 2)$, $\vec{v} = (3, 0, 1)$ and $\vec{w} = (1, -1, 2)$. If $\vec{t} = a\vec{u} + b\vec{v} + c\vec{w}$, then $a^2 + b^2 + c^2 =$

(a) 10

(b) 14

(c) 16

(d) 13

(e) 12

8. Let $S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_3 = 0\}$. Which one of the following is true

(a) S is neither closed under vector addition or closed under multiplication by scalar

(b) S is a subspace of $\mathbb{R}^3$

(c) The vector $(0, 1, 2) \in S$

(d) S is not closed under vector additions but closed under multiplication by scalar

(e) S is closed under vector addition but not closed under multiplication by scalar

9. Use the method of undetermined coefficients to find the general solution of the differential equation

$$4y'' + 4y' + y = 3xe^x$$

(No other method is accepted).

10. Use the method of variation of parameters to find a particular solution of the differential equation

$$y'' + 9y = 2\sec(3x).$$

11. Find a basis for the solution space of the given homogeneous system

$$\begin{cases} x_1 - 4x_2 - 3x_3 - 7x_4 = 0 \\ 2x_1 - x_2 + x_3 + 7x_4 = 0 \\ x_1 + 2x_2 + 3x_3 + 11x_4 = 0 \end{cases}$$

12. Find the general solution of the differential equation

$$(D-1)(D-2)^2(D^2+1)(D^2+6D+13)^2y=0$$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	Roots $2, \pm 2i$: $(m-2)(m^2+4) = m^3 - 2m^2 + 4m - 8$.
2	**(d)**	$\det = -3k - 1 \neq 0$, so $k \neq -\frac{1}{3}$.
3	**(a)**	Triangular determinant: $W = 1 \cdot 1 \cdot 2 = 2$.
4	**(a)**	y_c contains 1 and e^{-3x}, so multiply both trial groups by x.
5	**(b)**	$y = 5e^x + 2e^{3x}$, so $y(\ln 2) = 10 + 16 = 26$.
6	**(b)**	Row reduction leaves 3 nonzero rows, so the rank is 3.
7	**(b)**	Row reduce: $a = 2$, $b = -1$, $c = 3$, so $a^2 + b^2 + c^2 = 14$.
8	**(b)**	S contains $\mathbf{0}$ and is closed under addition and scalars: a subspace.
9	written	$y_p = (\frac{1}{3}x - \frac{4}{9})e^x$, so $y = c_1 e^{-x/2} + c_2 x e^{-x/2} + (\frac{1}{3}x - \frac{4}{9})e^x$.
10	written	$W = 3$, $u_1 = -\frac{2}{9}\ln\lvert\sec 3x\rvert$, $u_2 = \frac{2}{3}x$, so $y_p = -\frac{2}{9}\cos 3x \ln\lvert\sec 3x\rvert + \frac{2}{3}x \sin 3x$.
11	written	Row reduction leaves $x_3 = s$, $x_4 = t$ free; a basis is $\{\langle -1, -1, 1, 0\rangle, \langle -5, -3, 0, 1\rangle\}$.
12	written	Roots 1, 2 (double), $\pm i$, $-3 \pm 2i$ (double): $y = c_1 e^x + (c_2 + c_3 x)e^{2x} + c_4 \cos x + c_5 \sin x + (c_6 + c_7 x)e^{-3x}\cos 2x + (c_8 + c_9 x)e^{-3x}\sin 2x$.